

Antitank defenses

The German army quickly formulated antitank tactics after the first tank attack in September 1916. Artillery moved closer to the frontline: crews hid the guns and manhandled them into firing position should the enemy attack. The 77 mm field gun was converted to an antitank weapon by making the wheels smaller so the gun could be more easily hidden; trench mortars such as the 7.58cm Minenwerfer were given new mounts to enable them to be more easily fired at tanks, and a new 13 mm antitank rifle was put into production. Engineers dug hidden pits deep enough to stop a tank, and trenches were widened—8 ft-wide (2.5 m) was thought to be wide enough for the purpose. Another simple tactic was to bury artillery shells—again on likely approach routes—and place a pressure fuse in the shell. The round would then have a board placed over it to increase the pressure area: a charge of around 27–55 lb (12–25 kg) was considered enough to destroy a tank.

THE DOCTRINE OF MINES

Antitank mines in their hundreds of thousands were used in World War II. A mine didn't have to destroy a tank, only blow off or break a track: the crew would then have to either abandon the tank or try to repair it, making them vulnerable to machine gun fire and other weapons covering the minefield. Because of the threat of ditches, traps, and mines, various engineering vehicles were developed—such as these Combat Engineer Tractors—to overcome obstacles and allow an armored advance to continue.

Combat Engineer Tractors of the British 7th Armoured Brigade clear mines, January 7, 1991. Just over a week later they begin the liberation of Kuwait.

